

Question Paper

Exam Date & Time: 20-Mar-2024 (10:45 AM - 12:45 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

B.Tech IVth Semester Midterm Examination March 2024

DIGITAL SIGNAL PROCESSING [ECE 2222]

Marks: 30

Duration: 120 mins.

MCQ

Answer all the questions.

Section Duration: 20 mins

- 1) If the DFT of a real signal $x(n)$ is $X(k) = \{1, 6+j, Y, 4, 4+12j, 6-j\}$, then the value of Y is (0.5)
-

Correct option is: 2

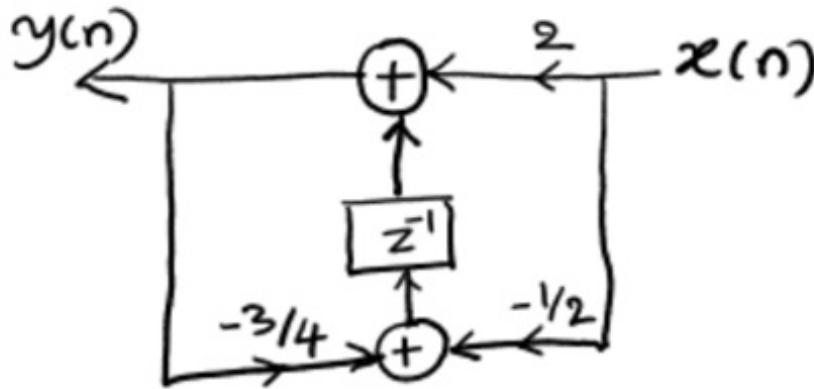
- 2) If $X(K)$ is the 8 point DFT of sequence $x(n)=[3,12,-7,0,9,-12,8,14]$ then the value of (0.5)

$$\sum_{k=0}^7 e^{\frac{-j3\pi k}{4}} X(K)$$

Correct option is: 2

- 3) (0.5)

Which of the following is the characterization for the system described by



- (a) $y(n) + \frac{3}{4}y(n-1) = 2x(n) - 0.5x(n-1)$
 (b) $y(n) - \frac{3}{4}y(n-1) = 2x(n) - 0.5x(n-1)$
 (c) $2y(n) + 0.5y(n-1) = x(n) + \frac{3}{4}x(n-1)$
 (d) $y(n) - 0.5y(n-1) = 2x(n) - \frac{3}{4}x(n-1)$

[\(a\)](#) [\(b\)](#) [\(c\)](#) [\(d\)](#)

Correct option is: 1

- 4) Lattice parameter (0.5)

K_1

of the lattice filter corresponding to the FIR filter described by the system function,

$$H(z) = 1 + 2z^{-1} + \frac{1}{3}z^{-2}$$

is

[1/3](#) [2/3](#) [3/2](#) [3](#)

Correct option is: 3

- 5) What is the roll-off rate of voltage gain vs frequency of a second order low pass filter? (0.5)

[6dB/octave](#) [12dB/octave](#) [6dB/decade](#) [2dB/decade](#)

Correct option is: 2

- 6) Consider the following statements regarding the impulse invariant method of IIR filter design (0.5)

- (a) After digitizing the analog filter, the impulse response of the original analog filter is preserved.
 (b) The method is inappropriate for the design of high-pass or band-stop filters. Which of these statements is/are correct?

[\(a\) only](#)
[\(b\) only](#)
[Both \(a\) and \(b\)](#)
[Neither \(a\) nor](#)

Correct option is: 3

- 7) Which of the following mapping is true between s-plane and z-domain? (0.5)
- | | | | |
|--|---|---|----------------------------------|
| Points in LHP of the s-plane into points inside the circle in z-domain | Points in RHP of the s-plane into points outside the circle in z-domain | Points on the imaginary axis of the s-plane into points onto the circle in z-domain | All of the above |
|--|---|---|----------------------------------|

Correct option is: 4

- 8) Which of the following filter transformation is not possible? (0.5)
- | | | | |
|--|---|---|-----------------------------------|
| High pass analog filter to low pass digital filter | High pass analog filter to high pass digital filter | Low pass analog filter to low pass digital filter | None of the above |
|--|---|---|-----------------------------------|

Correct option is: 2

- 9) Roll-off factor is (0.5)
- | | | | |
|---|---|---------------------------------|-----------------------------------|
| The bandwidth occupied beyond the Nyquist Bandwidth of the filter | The bandwidth of the filter | Aliasing effect | None of the above |
|---|---|---------------------------------|-----------------------------------|

Correct option is: 1

- 10) The nonlinear relation between the analog and digital frequencies is called____ (0.5)
- [Aliasing](#)
[Warping](#)
[Prewarping](#)
[Antialiasing](#)

Correct option is: 2

DESCRIPTIVE

Answer all the questions.

Section Duration: 100 mins

- 11) Determine the N-point DFT of the sequence (2)

$$x(n) = \begin{cases} 1, & 0 \leq n \leq \frac{N}{2} - 1 \\ 0, & \frac{N}{2} \leq n \leq N - 1 \end{cases}$$

- 12) Consider the following specifications for Goertzel algorithm (3)

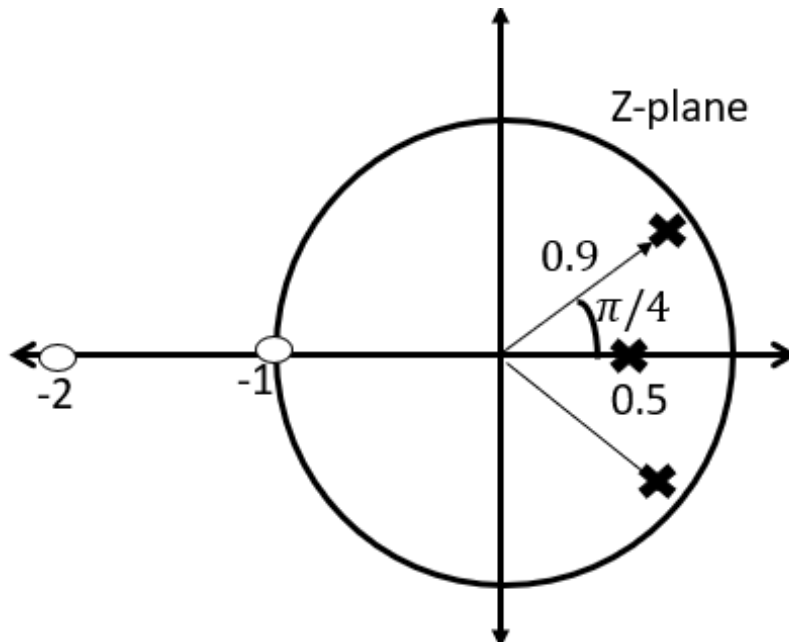
Signal frequency for detection: 7.5KHZ

DFT size: 32

Sampling frequency: 20KHZ

Determine the value of impulse response $h(n)$ and obtain the second order system function $H(z)$. Sketch the structure of the obtained system function $H(z)$.

- 13) Realize the system described by the following plot in direct form II by obtaining the system function followed by the linear constant coefficient difference equation. (3)



- 14) Consider an FIR filter with coefficients $K_1 = \frac{2}{3}$, $K_2 = \frac{3}{4}$, and $K_3 = \frac{1}{8}$. Determine the impulse response of the filter and realize the filter in direct form. (3)
- 15) Obtain the frequency sampling realization for the system whose frequency response is described by (3)
- $$H\left(\frac{2\pi k}{16}\right) = \begin{cases} 1, & k = 0, 1 \\ 0.5, & k = 2 \\ 0, & k = 3, 4, 5, 6, 7 \end{cases}$$
- 16) Transform the analog filter $H_a(s) = (S+1)/(S^2+5S+6)$ into $H(z)$ using impulse invariant transformation. (3)
Take $T = 0.1$ sec.
- 17) Develop N-point radix-2 DIT FFT algorithm with all intermediate mathematical expressions and diagrams. Illustrate with signal flow diagram (assume $N=8$). (4)
- 18) Design an analog Chebyshev type I lowpass filter that has 3dB cutoff frequency of 100 rad/sec and a stopband attenuation of 25 dB or greater for all the frequencies past 250 rad/sec. Determine the analog system function $H(s)$. (4)

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